

Creation and competition in the metaverse

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Abstract: The metaverse is a futuristic concept. It was first articulated in science fiction but has been pursued by corporations ever since. Due to technical limitations, the metaverse still lies in the future. Nevertheless, it is worthy of reflection today, also by legal scholars. The reason is that two arguably essential aspects of the metaverse, namely, interconnection between virtual worlds and a thriving creator economy, may never be achieved without legal intervention. In other words, whether the metaverse ever comes into existence depends not only on users and developers but also – even if more modestly – on lawyers. This paper shows how it starts by describing the metaverse, from its literary history to contemporary manifestations (Section 1), and then tackles two key issues, interoperability and mobile distribution, from a competition law and policy perspective (Section 2).

Keywords: Metaverse, Creator economy, Interoperability, Competition policy

1. What is the metaverse?

Like many futuristic concepts, the metaverse was first described in science fiction, in particular Neal Stephenson's 1992 novel *Snow Crash*.¹ Since then, different firms have tried to bring the metaverse into reality, each coming a little closer, but none of them attaining it quite yet. It is remarkable how closely these firms have stuck to Stephenson's vision. As Au puts it: 'Stephenson's Metaverse is less remembered as a science fiction novel than as a product roadmap'.² Given its outsized influence, *Snow Crash* serves as the red thread through my description of the metaverse, as I trace its evolution from the original idea to the virtual worlds in existence today (Section 1.1). A proper definition of the concept shows what is still needed to go from these virtual worlds to the metaverse (Section 1.2).

1.1. A short history of the metaverse: from *Snow Crash* to Meta

We first encounter the metaverse in the third chapter of *Snow Crash*.³ The novel's protagonist, Hiro Protagonist, is sitting down in his unit at the U-Stor-It. He puts on

a pair of goggles that wrap halfway around his head. A computer lens, connected to a fiber network, projects a 3D virtual world onto his goggles. He – or rather, his avatar⁴ – enters the metaverse, clad in a black leather kimono and packing two swords. The world he enters is a black sphere with a radius of about 10,000 km; a grand boulevard called The Street goes all the way around. It is populated by the 100 million richest people on earth, who use it daily for social, creative, commercial, and other pursuits.⁵

Snow Crash has the rare distinction of coining a term that is still in use 30 years later. The distinction is shared by Stephenson's fellow cyberpunk William Gibson, who coined the term 'cyberspace',⁶ which became synonymous with the web. However, while cyberspace sounds antiquated, the metaverse still lies in the future. At least in the literature, that future is dystopian: both Stephenson and Gibson describe ultra-violent, anarcho-capitalist worlds. Society is not doing well: Hiro escapes to the metaverse from the storage space in which he lives. A similar theme runs through Ernest Cline's *Ready Player One*, where people turn to the OASIS metaverse while society collapses around them.⁷

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1 Metaverse-like concepts were depicted in science fiction before; for an overview, see Matthew Ball, *The Metaverse: Building the Spatial Internet* (Liveright, 2024) 6–11.

2 Wagner James Au, *Making a Metaverse That Matters: From Snow Crash & Second Life to a Virtual World Worth Fighting for* (Wiley, 2023) 5.

3 Neal Stephenson, *Snow Crash* (1992; 2011 Penguin edn) 22 ('In the lingo, this imaginary place is known as the Metaverse').

4 The term 'avatar' was popularized, but not coined, by *Snow Crash*. As Stephenson acknowledges (Ibid 440), it was already used in its contemporary sense in the video game *Habitat*. For a description of avatars, see Ibid 33 ('audiovisual bodies that people use to communicate with each other in the Metaverse').

5 See Ibid 25 on the metaverse's population.

6 The term was popularized by William Gibson's *Neuromancer* (1984; 2016 Gollancz edn) but coined in his earlier short story *Burning Chrome*.

7 Ernest Cline, *Ready Player One* (Crown, 2011).

In these representations of 3D virtual worlds, video games play an important role.⁸ The first metaverse activity Stephenson mentions is building, but gaming comes not far behind.⁹ It is no surprise, then, that we caught the first glimpses of the metaverse in games, particularly those of id Software, the company that arguably invented 3D action gaming in the 1990s.¹⁰ In creating ever more persistent virtual worlds, developer John Carmack and his team were inspired by Stephenson's metaverse vision.¹¹ But their greatest contribution lay in licensing their game engines to other developers, who could use it to create their own virtual worlds.

At the turn of the millennium, the *Snow Crash* metaverse remained elusive. The web was compelling but, in the end, rather static and 2D, which spatial monikers like 'cyberspace' or 'information superhighway'¹² did not change. Video games were dynamic and 3D but lacked the mechanics of world-building and avatar personalization from Stephenson's metaverse, along with its massive scale. The closest thing to a metaverse took shape in Second Life, a world developed by Linden Labs, where users could – as the name suggests – lead a second life through a highly customizable avatar.¹³ Users could create virtual goods such as clothing and real estate, trade them for Linden dollars, and exchange those for US Dollars (USD).¹⁴

Second Life's creator economy, and its monetary integration with the real world, was a key development.¹⁵ By 2009, its economy had grown to over 500 million USD, with creators cashing out 55 million USD that year.¹⁶ But then Second Life stalled.¹⁷ While it had a loyal userbase, it was not able to attract and retain a wider audience. Thus, it never reached the massive scale that characterizes the metaverse. The mistake, according to Second Life developer Richard Nelson, was 'not launching as a game first'.¹⁸ Epic Games founder Tim Sweeney agrees: '[I]t's not very fun to sit around in 3D and just talk to people'.¹⁹ Other metaverse hopefuls, including Sweeney, would launch their virtual world as a game first.

Minecraft was largely developed by Markus Persson (known as 'Notch'). He released Minecraft in 2009 with two modes: a Survival mode, typical of games, but also a Creative mode, where players can use blocks to create fantastic structures.²⁰ By 2023, with 300 million copies sold, it had become the best-selling video game ever.²¹ Other 'sandbox' games, where creation by users is central, cannot beat that sales record because they are not sold. Both Fortnite, by Epic Games, and Roblox are free-to-play games with creative modes. In Roblox Studio, for example, users can create 'anything they can imagine'²² and then share their creations with the platform's 80+ million daily active users.²³ Calling Minecraft, Roblox

8 See Gibson (fn 6) 59 ('The matrix has its roots in primitive arcade games'). In Cline *Ready Player One* (fn 7), the OASIS started as a game but then evolved into a general-purpose world (education being a featured use case).

9 See Stephenson (fn 3) 23–24 on building and 36 on video games (in the form of metaverse amusement parks that offer 'interactive three-dimensional movies').

10 For games like Wolfenstein 3D, Doom, and Quake, see Michel Mariott, 'In the Background, a Man in the Gaming Forefront' *New York Times* (5 August 2004), available at: <https://nytimes.com/2004/08/05/technology/in-the-background-a-man-in-the-gaming-forefront.html>.

11 David Kushner, *Master of Doom: How Two Guys Created an Empire and Transformed Pop Culture* (2003; Piatkus, 2004 edn) 189–90 (on Quake) and 282–83 (on Carmack pitching a more generalized infrastructure of which games would only form part).

12 The term was popularized by Al Gore and later by Bill Gates, *The Road Ahead* (Viking, 2005).

13 On Second Life generally, see Au (fn 2) ch 2.

14 On Second Life's currency system, see <https://community.secondlife.com/knowledgebase/english/buying-and-selling-linden-dollars-r46/>.

15 Contemporary games, such as World of Warcraft and RuneScape, also had developed in-game economies. However, their developers did not allow currency trading, which meant that any integration with the real-world economy was informal. On game economies, see Isaac Knowles, Edward Castronova and Travis Ross, 'Video games, virtual worlds and economics' in Robert Picard and Steven Wildman (eds), *Handbook on the Economics of the Media* (Edward Elgar, 2015).

16 Dean Takahashi, 'Second Life's economy grows 65% to \$567M' (*VentureBeat*, 19 January 2010), available at: <https://venturebeat.com/business/second-lifes-economy-grows-65-to-567m/>.

17 Even as user numbers stalled, the Second Life economy kept running. By 2024, Linden Labs had paid 1.1 billion USD to creators; see Dean Takahashi, 'Linden Lab has spent \$1.3B building Second Life and paid \$1.1B to creators' (*VentureBeat*, 19 December 2024), available at: <https://venturebeat.com/games/linden-lab-has-spent-1-3b-building-second-life-and-paid-1-1b-to-creators/>.

18 Au (fn 2) 62.

19 Patrick McGee, 'Tim Sweeney: epic will fight Apple and Google to keep the metaverse open' *Financial Times* (26 May 2022), available at: <https://ft.com/content/e13ce526-0e33-4ca2-9699-184d0138eada>.

20 See Edwin Evans-Thirlwell, *The World of Minecraft: A Visual History of the Global Phenomenon* (Expanse, 2024) 14. Creative mode was launched first but, as Notch noted (in line with Nelson and Sweeney): 'Free-building mode is fine and dandy ... But for many people it will become boring once you've got it figured out'. Hence, he inserted more traditional gameplay elements (survival), before later spinning off creative mode.

21 Tom Gerken, 'Minecraft becomes first video game to hit 300m sales' (*BBC*, 16 October 2023), available at: <https://www.bbc.com/news/technology-67105983>.

22 See <https://create.roblox.com/>. Fortnite has a similar Creative mode, see <https://dev.epicgames.com/community/fortnite/getting-started/creative>.

23 Roblox Corporation, Annual report (Form 10-K) for the fiscal year ended 31 December 2024, submitted to the U.S. Securities and Exchange Commission, 9, available at: <https://ir.roblox.com/financials/sec-filings/default.aspx>. Questioning some of Roblox's metrics (and its content moderation), see 'Roblox: inflated key metrics for Wall Street and a pedophile hellscape for kids' (*Hindenburg Research*, 8 October 2024), available at: <https://hindenburgresearch.com/roblox/>. Fortnite does not release daily active user numbers but reports concurrent users, which hover between one and three million, see <https://fortnite.gg/player-count>.

and Fortnite 'games' has become reductive, as they have evolved into wider spaces not only for entertainment²⁴ but also for education²⁵ and even politics.²⁶

In 2021, Mark Zuckerberg symbolically tied Facebook's fate to the metaverse by renaming the firm Meta.²⁷ With two billion daily active users, Facebook represented the metaverse's scale more than any other platform. However, social networks were based on users' real-world identities and lacked the spatial nature of the metaverse. This would change as Meta turned its focus to providing 'an embodied internet where you're in the experience' as an avatar – not just looking at it on the screen.²⁸ The seeds of this transformation had been sown in 2014, when Meta acquired virtual reality (VR) headset maker Oculus, a firm that already set its sights on the metaverse.²⁹ At the time, Zuckerberg stated: 'Mobile is the platform of today, and now we're also getting ready for the platforms of tomorrow.'³⁰ He held onto that narrative in 2021, when he described the metaverse as 'the successor to the mobile internet.'³¹ This grand vision is also driven by business strategy: having been forced to live in the mobile ecosystems of Apple and Google, Meta is keen to be in control of its destiny in the next platform cycle.³² It is certainly willing to spend to make that happen, to the tune of 65 billion USD in the period 2019–2024.³³ So far, this has not translated into significant user engagement in Meta's main virtual world, Horizon Worlds.³⁴

Note, finally, that in regions outside Europe and North America, other platforms are equally or more important. Fortnite has a gaming-focused metaverse competitor in Free Fire.³⁵ The Korean metaverse platform Zepeto, owned by Naver, skipped gaming by starting as a social platform.³⁶ A type of metaverse can also be

discerned in the super-apps that are popular in parts of Asia. In Tencent's WeChat, for example, profiles can be personalized, and 'mini apps' support a wide variety of experiences.³⁷

1.2. The metaverse defined: core and periphery

The metaverse came a long way from its science fiction origins to its most recent manifestations. To know whether we have reached the metaverse, however, we need a clear definition. The definitional question is complicated by the different terms that are floating around: while Meta sticks to 'metaverse', Apple prefers 'spatial computing', Tencent's 'hyper digital reality' and Roblox's 'human co-experience'.³⁸ Nevertheless, there are characteristics that unite most visions. The metaverse is (i) immersive (or spatial or 3D), allowing users to exist within it in the form of an avatar; (ii) continuous, which means users have ongoing access to their identity, entitlements, items, communications, and payments; (iii) persistent, or permanent, whether users are present or not; (iv) synchronous, which implies that – when present – users' experience is or can be shared in real time; and (v) massively scaled, in the way that the internet is.³⁹

Each of these features was present in *Snow Crash*,⁴⁰ but no platform combines all of them today. The virtual worlds that grew out of sandbox games come closest but, due to technical limitations (related to networking, compute, and code), cannot yet synchronously render enough users to mirror *Snow Crash*'s massive scale.⁴¹ Fortnite's Travis Scott concert may have been attended by 12 million users 'at the same time', but those users were divided

24 Travis Scott's Fortnite shows drew a total of 46 million viewers, Lil Nas X's Roblox shows 33 million, see Jacob Kastrenakes, 'Lil Nas X's Roblox concert was attended 33 million times' (*The Verge*, 16 November 2020), available at: <https://theverge.com/2020/11/16/21570454/>.

25 Minecraft in particular is used in classrooms, see <https://education.minecraft.net/en-us/discover/what-is-minecraft>.

26 Kim Lyons, 'The Biden-Harris campaign has launched a "Build Back Better" map in Fortnite' (*The Verge*, 31 October 2020), available at: <https://theverge.com/2020/10/31/21543360/>.

27 Mark Zuckerberg, Keynote (*Meta Connect*, 28 October 2021), available at: <https://facebook.com/RealityLabs/videos/561535698440683/>. For a transcript, see <https://rev.com/blog/transcripts/meta-facebook-connect-2021-metaverse-event-transcript>.

28 Ibid, referring to 'a shared sense of space' or 'feeling of presence' as crucial.

29 See the pre-acquisition blog post 'Onward to the Metaverse!' (*Meta Quest Blog*, 14 December 2013), available at: <https://meta.com/en-gb/blog/quest/onward-to-the-metaverse/>.

30 'Facebook to acquire Oculus' (*Meta Newsroom*, 25 March 2014), available at: <https://about.fb.com/news/2014/03/facebook-to-acquire-oculus/>.

31 Zuckerberg (fn 27).

32 Matthew Ball, 'Interviewing Meta CTO Andrew Bosworth on the metaverse, VR/AR, AI, billion-dollar expenditures, and investment timelines' (9 July 2024), available at: <https://matthewball.co/all/bozinterview2024>.

33 Ibid.

34 Jeff Horwitz, Salvador Rodriguez and Meghan Bobrowsky, 'Company documents show meta's flagship metaverse falling short' (*Wall Street Journal*, 15 October 2022), available at: <https://wsj.com/articles/meta-metaverse-horizon-worlds-zuckerberg-facebook-internal-documents-11665778961>.

35 The battle royale game also features a creative mode, see <https://ffsupport.garena.com/hc/en-us/articles/4412913818522-Craftland-Feature>.

36 Christian Davies and Song Jung-a, 'Asia's largest metaverse platform Zepeto ramps up global expansion' (*Financial Times* (27 September 2022)), available at: <https://www.ft.com/content/14c88e84-f3c8-485e-a9df-31ead34e48f0>.

37 Yilun Chen, *Influence Empire: The Story of Tencent and China's Tech Ambition* (Hodder & Stoughton, 2022) 106–109.

38 Matthew Ball, 'On spatial computing, metaverse, the terms left behind and ideas renewed' (6 February 2024), available at: <https://matthewball.co/all/metaversespatialandmore>.

39 Ball (fn 1) ch 4; Fabiana Di Porto, Daniel Foà and Sean Ennis, 'Emerging virtual worlds: implications for policy and regulation' (CERRE Report, 2024) 16–17.

40 The internet-like scale was not explicitly present, as Stephenson wrote about only 100 million daily active users, which has been far exceeded by the internet. The internet did serve as the benchmark for that number, but it was unclear at the time how large it would become.

41 On those limitations, see Ball (fn 1) chs 5–7.

into groups of 50 ('shards'), and their functionality was limited.⁴² That is one reason why Fortnite and similar virtual worlds can at most be called 'proto-metaverses'.

There is another reason we only have proto-metaverses, and that is because the above definition leaves out two key characteristics that are currently not or hardly met.⁴³ The first is interoperability. The metaverse is not just a virtual world; rather, it consists of different virtual worlds that are interconnected. Such interoperability is lacking today, even though firms and policymakers are working toward it, as I discuss in Section 2.1. The second key characteristic is that the metaverse must have powerful user creation tools that are at the basis of a vibrant virtual economy integrated with the real world. Some virtual worlds already facilitate significant user creation and monetization. Roblox is a good example. Its users can create experiences (often games), avatar items, and developer tools. They can then make money from their creations: items and tools are sold; experiences are monetized via in-experience purchases, engagement-based payouts, or ad impressions. In 2024, over five million Roblox users earned in-game currency – Robux – for their creations; 17,000 cashed out their Robux for a total of 920 million USD.⁴⁴ But for creators to thrive, online and offline, mobile distribution may need to change, as I discuss in Section 2.2.

In addition to these core aspects of the metaverse, there is the periphery: the phenomena that are often spoken about in the same breath as the metaverse but are not essential. The first one is VR (along with its brethren, augmented and mixed reality). Headsets like the (Oculus) Quest are just one way of accessing the metaverse, as Meta itself recognized when it rolled out its Horizon Worlds on the web and mobile.⁴⁵ *Snow Crash's* metaverse can similarly be accessed via terminals, although they degrade the

experience.⁴⁶ Then, there are enabling technologies: 'Web 3.0' technologies like blockchain can enable interoperability, and artificial intelligence can help procedurally generate worlds, but neither is strictly necessary.⁴⁷

2. Competition and creator issues

The metaverse, like the virtual worlds that precede it, will run into legal issues relating to intellectual property (IP),⁴⁸ crime,⁴⁹ and finance⁵⁰ – to name just a few. I choose to focus on two issues – interoperability and mobile distribution – that are existential, as in required for the metaverse to even come into being.⁵¹ Both relate to competition, and both concern creators.

From a competition standpoint, it is useful to complement the metaverse's definition with an overview of its value chain. I distinguish four layers.⁵² First, there is the infrastructural layer, which includes physical infrastructure (e.g., internet network, chips) but can be extended to foundational digital services (e.g., game engines). Second, there is the platform layer, which consists of the virtual worlds that host user-generated content. Third, there is the application layer, where creators develop the experiences that populate the platform. Fourth, there is the interface layer, which is how users connect to the metaverse, be it via a headset or the screen of a smartphone or computer.

Some of these layers are more concentrated than others. They can even be monopolized. In *Snow Crash*, the fiber network powering the metaverse is fully in the hands of L Bob Rife.⁵³ Firms can also be integrated across layers. Epic Games operates Fortnite, a platform, but also the Unreal Engine, the game engine of choice for demanding graphics.⁵⁴ Often, vertical integration

42 Ibid 117 and 130.

43 For references on their characteristics and their essential nature, see Sections 2.1 and 2.2.

44 Roblox Corporation (fn 23) 9–11 and 16–17. By way of comparison, Fortnite had 70,000 creators earning a total of 350 million USD in 2024, see 'Fortnite ecosystem 2024 year in review: celebrating creators and looking ahead' (*Epic Games Creator Portal*, 22 January 2025), available at: <https://create.fortnite.com/news/fortnite-ecosystem-2024-year-in-review-celebrating-creators-and-looking-ahead>.

45 Amanda Silberling, 'Meta's Horizon Worlds is rolling out on web and mobile' (*TechCrunch*, 14 September 2023), available at: <https://techcrunch.com/2023/09/14/metahorizon-worlds-is-rolling-out-on-web-and-mobile/>.

46 Stephenson (fn 3) 38 (distinguishing between 'cheap public terminals' that render the user in grainy black-and-white and 'pay terminals' with better image quality).

47 See Ball (fn 1) 27–30 and 31–33, respectively.

48 See, e.g., Gaetano Dimita, Yin Harn Lee, Michaela MacDonald, Anthony Michael Catton, Zeynep Kubra Kavcar Penbegullu and Juan Alberto Pulido Lock, 'IP and metaverse(s)' (Research Report commissioned by Intellectual Property Office, March 2024).

49 See, e.g., Eldar Haber, 'The criminal metaverse' (2024) 99 *Indiana Law Journal* 843.

50 See, e.g., Hung-Yi Chen, Pawee Jenweeranon and Nafis Alam (eds), *Global Perspectives in the Metaverse: Law, Economics, and Finance* (Palgrave Macmillan, 2024) in particular chs 8–11.

51 This is not to say the metaverse *needs* to come into being, though I believe it to be a neutral concept: the science fiction worlds with metaverses may be dystopian, but their metaverses are not (though one may worry that focusing on virtual worlds takes our attention away from issues in the real world).

52 Based on 50 submissions to its call for contributions, the Commission distinguishes three layers: enabling technologies, platforms, and services, see Klaus Kowalski, Cristina Volpin and Zsolt Zombori, 'Competition in generative AI and virtual worlds' (European Commission Competition Policy Brief, September 2024) 10–11. A popular account distinguishes seven layers, see Jon Radoff, 'The metaverse value-chain' (*Meditations*, 7 April 2021), available at: <https://meditations.metavert.io/p/the-metaverse-value-chain-afc9e09e3a7>.

53 Or *almost* fully, as Rife regrets that a monopolist 'can never get that last one-tenth of one percent', see Stephenson (fn 3) 106–107.

54 2D, mobile, and indie games tend to use Unity Technologies' Unity engine, but some metaverse platforms such as VRChat also do, see <https://creators.vrchat.com/sdk/upgrade/current-unity-version/>.

proceeds through acquisition. While Meta mainly operates a platform, its purchase of Oculus gave it an interface. It later strengthened its foothold in the app layer with the acquisition of VR fitness app developer Within.⁵⁵ Microsoft has made a number of gaming acquisitions. It bought Minecraft developer Mojang for 2.5 billion USD in 2014 and ZeniMax for 7.5 billion USD in 2020.⁵⁶ In 2022, it announced its 69 billion USD acquisition of Activision–Blizzard, which would provide it with ‘building blocks for the metaverse’.⁵⁷

Acquisitions can raise competition concerns, both horizontally and vertically. Horizontal acquisitions (in the same market) can create or strengthen market power, which can then be exploited.⁵⁸ A metaverse platform that gains (increased) market power could, for example, use it to reduce payments to creators.⁵⁹ Vertical integration can change a firm’s incentives: post-acquisition, it may want to withhold or degrade access to its content (input foreclosure) or distribution channels (customer foreclosure).⁶⁰ If the content or distribution channels in question are significant to competitors, such post-merger conduct can distort competition. A metaverse platform (or interface) could, for example, refuse to distribute competing content or could deny its own ‘must-have’ content to other platforms.⁶¹

The focus of this article is not on merger control but on unilateral conduct, and in particular on the issues of interoperability and mobile distribution, though the competition concerns – exploitation and exclusion (foreclosure) – are similar. In my discussion of interoperability and mobile distribution, Roblox will (continue to) serve as a case study for three reasons. First, with its 80+ million daily active users, Roblox is the largest proto-metaverse – and it continues to grow.⁶² Second, of all proto-metaverses, Roblox is closest to the ‘real thing’: it is not a game itself (as Fortnite is) but rather an empty world populated entirely by user-created experiences. Third, as

a public company, Roblox’s financials are public, which is helpful to pinpoint economic issues and gauge their magnitude, especially when it comes to creator remuneration.

2.1. Interoperability

In *Snow Crash* and *Ready Player One*, there is only one metaverse. Commentators have taken this description as normative. Tony Parisi has formulated seven rules for the metaverse, the first one being: ‘There is only one Metaverse’.⁶³ In science fiction worlds, however, the singular metaverse was a natural outcome: because these worlds were either the first or the best, users were drawn to Stephenson’s metaverse or Cline’s OASIS until they all ended up there. In economic terms, the market ‘tipped’ to a winner-takes-all outcome.⁶⁴ In the real world, a singular metaverse is unlikely to be a natural outcome. Hence, the many virtual worlds would need to be made interoperable to realize the vision of a single metaverse.⁶⁵

In this section, I examine the desirability of and options for metaverse interoperability. I first tackle the grand vision of a single, interoperable metaverse (Section 2.1.1). Next, I take a more granular look at different forms of interoperability – of communications (Section 2.1.2), identities (Section 2.1.3), social graphs (Section 2.1.4), assets (Section 2.1.5), and currencies (Section 2.1.6) – before concluding (Section 2.1.7). Along the way, I point to the trade-offs presented by interoperability and the often limited powers under competition policy to mandate it.

2.1.1. An interoperable metaverse

The interoperability requirement that some impose on the metaverse is supported by internet comparisons. Zuckerberg and others have described the metaverse as ‘the next version of the internet’.⁶⁶ The internet is

55 The Federal Trade Commission (FTC) tried but failed to block the acquisition, see *Fed. Trade Comm’n v. Meta Platforms Inc.*, 654 F. Supp. 3d 892 (N.D. Cal. 2023).

56 The European Commission approved the acquisition, see *Microsoft/ZeniMax* (Case M.10001) Commission Decision of 5 March 2021.

57 ‘Microsoft to acquire Activision Blizzard’ (*Microsoft News Center*, 18 January 2022), available at: <https://news.microsoft.com/2022/01/18/microsoft-to-acquire-activision-blizzard-to-bring-the-joy-and-community-of-gaming-to-everyone-across-every-device/>. Competition authorities in the EU, UK, and elsewhere scrutinized the deal, but it finally went through. In the EU, see *Microsoft/Activision Blizzard* (Case M.10646) Commission Decision of 15 May 2023. In the UK, the Competition & Markets Authority (CMA) initially blocked the acquisition, see Anticipated acquisition by Microsoft of Activision Blizzard, Inc. (Final Report, 26 April 2023), but eventually approved a restructured deal with commitments.

58 See ‘Guidelines on the assessment of horizontal mergers under the council regulation on the control of concentrations between undertakings’ [2004] OJ C31/5.

59 By analogy, see *United States v. Bertelsmann SE & Co. KGaA*, 646 F. Supp. 3d 1 (D.D.C. 2022), where a U.S. court prohibited a merger between two

major book publishers because it would lead to reduced advances for authors.

60 ‘Guidelines on the assessment of non-horizontal mergers under the council regulation on the control of concentrations between undertakings’ [2008] OJ C265/8.

61 In the *Microsoft/Activision Blizzard* decisions (fn 57), competition authorities examined both risks in a gaming context. Given that this was primarily a content acquisition, input foreclosure concerns (e.g., of the must-have Call of Duty) were more prominent.

62 See Section 1.1.

63 Tony Parisi, ‘The seven rules of the metaverse’ (*Medium*, 22 October 2021), available at: <https://medium.com/meta-verses/the-seven-rules-of-the-metaverse-7d4e06fa864c>.

64 Geoffrey Parker, Marshall Van Alstyne and Sangeet Choudary, *Platform Revolution* (Norton, 2016) 224–27.

65 Compare Ball (fn 1) 102–105 (considering the metaverse an interoperable network), with Au (fn 2) 194–200 (calling it a myth that the metaverse must interoperate across the internet).

66 Zuckerberg (fn 27).

famously not one network but a network of networks that interoperate based on the internet protocol suite (TCP/IP).⁶⁷ The *Snow Crash* metaverse also runs on a protocol, governed by the Global Multimedia Protocol Group⁶⁸ – a parody of the Internet Engineering Task Force (IETF), which maintains the internet protocol suite.⁶⁹

The vision of an open, interoperable metaverse is utopian. The background is that the internet was architected to be open but did not remain so. In the 1990s, online service providers like AOL tried to keep users in their walled gardens, which were largely separated from the open internet.⁷⁰ But the freedom of the internet – and its killer app, the web – won out. The open internet was less resistant to the rise of closed platforms in the 2000s. Internet protocols were not designed to run complex services such as search and e-commerce in an interoperable way.⁷¹ The shift from desktops to more closed mobile devices, which are even more app- than web-based, sealed the open internet's fate.⁷² Now, the metaverse is rekindling hopes of an open online space.

Firms and policymakers are unusually aligned on the metaverse's openness. The European Commission ties its metaverse policy to the evolution of the internet by discussing it as part of 'web 4.0'.⁷³ It defines the metaverse as an '[i]nteroperable network of virtual worlds,' stressing the need for open standards.⁷⁴ This is aligned with the views of Meta. According to Zuckerberg, the metaverse requires interoperability so that 'you're not going to be locked into one world or platform'.⁷⁵ Or, as its head of EU affairs put it: 'We don't have several internets, that's why we want to have one metaverse'.⁷⁶ Sweeney agrees the metaverse is about connecting different networks based on open standards – 'exactly how the internet came about'.⁷⁷

There are reasons to be skeptical, some more practical, others more fundamental. EU policy remains high level

and may underestimate the standardization needed to bring about the envisaged interoperability. For firms like Meta, interoperability is a matter of business strategy, in line with the lesson that incumbents are closed and challengers pursue openness.⁷⁸ Abstracting from the interests at play, the desirability of interoperability policies – and the standardization required to put it into practice – depends largely on two criteria: the strength of network effects and the heterogeneity of user demand. Put simply: if the value of the network increases with the number of users, standardization is beneficial; if users have different preferences, it is not.⁷⁹

When applied to the metaverse, the two criteria pull in different directions. On the one hand, if the metaverse becomes a place where people lead a 'second life,' they clearly benefit from having others (at least those they care about) 'live' in the same virtual world. On the other hand, people likely have wildly different ideas about what their preferred world looks like, freed from the constraints of the real one. To complicate matters, those preferences are not constant: people might want to spend their morning in one world and their afternoon in another.

There are two ways for interoperability policies to navigate the trade-off between the breadth of the network and its diversity. The first way is to standardize the underlying infrastructure while allowing for a variety of experiences on top of it. The internet does this to some extent though, as noted, modern services do not run on the internet protocol suite. I return to this option in the conclusion (Section 2.1.7). The second way is to recognize that interoperability comes in different forms and degrees. Interoperability between virtual worlds can be seen as the metaverse's end-state, but there are important steps in between. I distinguish five types of interoperability that are more technically and legally feasible. They concern communications, identity, social graphs,

67 On how the TCP/IP protocol came to be and won out against other protocols, see Katie Hafner and Matthew Lyon, *Where Wizards Stay Up Late: The Origins of the Internet* (1996; 2006 Simon & Schuster edn) 246–54.

68 Stephenson (fn 3) 23–24.

69 On IETF and related standard-developing organizations, see Alison Harcourt, George Christou and Seamus Simpson, *Global Standard-Setting in Internet Governance* (Oxford University Press, 2020) ch 3.

70 Brian McCullough, *How the Internet Happened: From Netscape to the iPhone* (Liveright, 2018) 62–64.

71 Some governments are now trying to change that. India, for example, launched the Open Network for Digital Commerce, which enables e-commerce based on an open protocol, see <https://www.ondc.org/>.

72 Apple's iOS is largely closed, though its App Store provides an entryway. Google's Android is nominally open but has been closing down over the years, see Ron Amadeo, 'Google's iron grip on Android: Controlling open source by any means necessary' (*Ars Technica*, 20 October 2013 [updated 21 July 2018]), available at: <https://arstechnica.com/gadgets/2018/07/googles-iron-grip-on-android-controlling-open-source-by-any-means-necessary/>.

73 European Commission, 'An EU initiative on Web 4.0 and virtual worlds: a head start in the next technological transition' (Staff Working Document) SWD(2023) 250.

74 Ibid 86.

75 Zuckerberg (fn 27).

76 Aura Salla (now a member of the European Parliament) on LinkedIn (2023), available at: https://www.linkedin.com/posts/aura-salla-8ab84326_meta-metaverse-innovation-activity-7010870388838563840-aFZN/ (accessed 28 October 2024 (now no longer publicly available)).

77 Andrew Webster, 'Tim Sweeney explains how the metaverse might actually work' (*The Verge*, 23 March 2023), available at: <https://theverge.com/2023/3/23/23652928/>.

78 Michael Katz and Carl Shapiro, 'Network externalities, competition, and compatibility' (1985) 75 *The American Economic Review* 424, 425. See Alex Heath, 'Zuckerberg says Meta and Apple are in "very deep, philosophical competition" to build the metaverse' (*The Verge*, 26 July 2022), available at: <https://theverge.com/2022/7/26/23279478/> ('Meta would position itself as the more open ... alternative to Apple').

79 Christopher Yoo, *The Dynamic Internet: How Technology, Users, and Businesses Are Transforming the Network* (AEI Press, 2012) 74–78.

items, and currencies. Each can be seen as a step toward an interoperable metaverse, though they need not follow this specific order. Let us discuss each type of interoperability in turn.

2.1.2. Communications interoperability

Communications or, more restrictively, messaging interoperability is the most well-known type of interoperability. Email services have always been interoperable. Thanks to the internet's Single Mail Transfer Protocol (SMTP), nothing prevents communication between Gmail and Outlook.⁸⁰ Public authorities have sought to maintain interoperability as means of communication evolved. In 2000, AOL was the dominant instant messaging (IM) provider. As it benefitted from significant network effects, observers expected that to remain the case. When AOL merged with Time Warner, the Federal Communications Commission therefore required AOL to ensure interoperability between its IM service and competitors.⁸¹ This would 'continue the long-standing tradition of the Internet being open and interoperable' and emulate 'the healthy competition that exists today in electronic mail'.⁸²

Messaging standards have remained contentious since. Google has long tried to push Apple to move on from the Short Message Service (SMS) to the Rich Communication Services (RCS) protocol so that messages between iOS and Android users could include modern features.⁸³ The Digital Markets Act (DMA) obliges designated gatekeeper services to make their communications services interoperable.⁸⁴ As iMessage was not

designated,⁸⁵ Apple need not abide by the obligation, though it has meanwhile announced it will implement RCS.⁸⁶ Meta's WhatsApp and Messenger have been designated as gatekeeper services,⁸⁷ so the firm is making them interoperable with other communications services.⁸⁸

Communications services provide a good case study to further discuss the downsides of interoperability. First, there is the technical difficulty. Communications, and certainly messaging, services are relatively simple apps, but making even them interoperate is challenging.⁸⁹ The DMA recognizes this, as it gives gatekeepers four years to implement voice and video call interoperability.⁹⁰ Second, there is the privacy and security risk. End-to-end encryption (E2EE) is the gold standard for messenger services, but once they have to interoperate, it becomes virtually impossible to guarantee E2EE without far-reaching standardization on the same protocol.⁹¹ To interoperate securely, Meta recommends that connecting services use the Signal Protocol, but it also allows others.⁹² Third, standardization can reduce innovation. Signal founder Moxie Marlinspike has long warned that protocols evolve slowly, if at all: SMTP, for example, is stuck in the 1990s.⁹³ Moreover, standards tend to be led by the dominant player or – perhaps worse – represent the lowest common denominator of all players, which means that all services may end up looking alike. This limits innovation in standards-based messenger services.

These downsides should give policymakers pause when mandating interoperability. The objections may not be disqualifying for communications interoperability: messaging is in any case not the most disruptive service, and protocols already exist. However, the objections

80 For that reason, the email services of Microsoft and Google were not designated as gatekeepers under the DMA, see *Microsoft* (Case DMA.100023) Decision of 5 September 2023, para 128 and *Alphabet* (Case DMA.100006) Decision of 5 September 2023, paras 136 and 144.

81 *Time Warner Inc. & Am. Online, Inc.*, Memorandum Opinion & Order, FCC No. 01-12 (2001) paras 191–200. AOL was not allowed to introduce more advanced applications (in particular videoconferencing) before it had implemented 'a standard for server-to-server interoperability of [Names and Presence Database]-based services that has been promulgated by the IETF or a widely recognized standard-setting body' (para 192).

82 Ibid paras 131 and 163. Two and a half years later, the FCC removed the condition, see, *Petition of AOL Time Warner Inc. for Relief from the Condition Restricting Streaming Video AIHS*, FCC, No. 03-193 (2003).

83 See, e.g., Jon Porter, 'Google turns to regulators to make Apple open up iMessage' (*The Verge*, 8 November 2023), available at: <https://theverge.com/2023/11/8/23951935/>.

84 'Regulation (EU) 2022/1925 of the European Parliament and of the Council on contestable and fair markets in the digital sector' [2022] OJ L265/1 (hereafter: DMA) Art. 7.

85 *Apple – number-independent interpersonal communication services* (Case DMA.100022) Commission Decision of 12 February 2024.

86 Allison Johnson, 'Apple is Bringing RCS to the iPhone in iOS 18' (*The Verge*, 10 June 2024), available at: <https://theverge.com/2024/6/10/24171315/>.

87 *Meta* (DMA.100024) Commission Decision of 5 September 2023, ss. 5.4 and 5.5.

88 Meta, 'Non-confidential public summary of Meta's Compliance Report' (6 March 2025) paras 147–77, available at: <https://transparency.meta.com/reports/regulatory-transparency-reports/>.

89 Meta making its own communications services interoperable involved 'thousands of Facebook employees to reconfigure how WhatsApp, Instagram and Facebook Messenger function at their most basic levels', see Mike Isaac, 'Zuckerberg plans to integrate WhatsApp, Instagram and Facebook Messenger' *New York Times* (25 January 2019), available at: <https://nytimes.com/2019/01/25/technology/facebook-instagram-whatsapp-messenger.html>. Five years later, the integration had not been completed.

90 DMA, Art. 7(2)(c). Other obligations must be implemented within six months after designation.

91 Ibid Art. 7(3) obliges gatekeepers to preserve the level of security, including E2EE. Without a common protocol, however, messages have to be re-encrypted. And in any case, you end up trusting the other system to keep the messages safe. Interoperability increases the attack surface of conversations. See Matthew Hodgson, 'How do you implement interoperability in a DMA world?' (*Matrix Blog*, 29 March 2022), available at: <https://matrix.org/blog/2022/03/29/how-do-you-implement-interoperability-in-a-dma-world/>.

92 Meta (fn 87) para 165.

93 Moxie Marlinspike, 'Reflections: the ecosystem is moving' (*Signal Blog*, 10 May 2016), available at: <https://signal.org/blog/the-ecosystem-is-moving/>. For a nuanced discussion, see Matthew Wild, 'Products vs protocols: what signal got right' (*Snikket Blog*, 9 February 2021), available at: <https://snikket.org/blog/products-vs-protocols/>.

gain in strength as the services at issue become more differentiated. The market should not be underestimated either. In gaming, Discord evolved into a near-universal communications app. After its founding in 2015, players increasingly used it as an alternative to the communications services provided by platforms (e.g., consoles) and games.⁹⁴ Most platforms and many games ended up integrating Discord,⁹⁵ making interoperable communications a reality.⁹⁶

2.1.3. Identity interoperability

It would be valuable to be able to take your digital identity from one virtual world to another.⁹⁷ Users would not have to start an account from scratch. It may even offer a way to protect users and creators from imposters in other worlds. A minimal level of interoperability already exists in the ‘log in with [Meta/Google/Apple]’ features. Many virtual worlds have integrated application programming interfaces (APIs) of Meta, Apple, and Google so that users can log in via those services, taking basic personal information with them.

But this system can raise competition concerns.⁹⁸ When Meta required users of its VR headsets to register with a Facebook account, the German competition authority started investigating. The authority believed this requirement would tie users even more closely to Meta’s ecosystem. It also objected to Meta’s linking of data between users of both its social network and VR headset. Both practices would raise barriers to entry for new players. The investigation ended when Meta allowed users to set up a separate Meta Quest account and committed not to combine data between services.⁹⁹ Inspired by the Bundeskartellamt’s investigation and others like it, the DMA regulates the linking of services. Gatekeepers cannot oblige end-users to connect with their identification service or require business users to interoperate with it.¹⁰⁰ It also limits the extent to which gatekeepers can combine data across services.¹⁰¹

The dominant/gatekeeper firms that are best placed to come up with an interoperable log-in system are thus limited in doing so. They are not prohibited but need to take extra care to respect user freedom. Perhaps there is room for an independent player like Discord to provide universal identification. An independent player (or organization) is more likely to be trusted by business users, as – unlike large tech firms – it does not compete with them in providing a virtual world.¹⁰² Even then, any system will face challenges, technical and social, especially once interoperability is extended from simple identification to avatars. Limitations have to be built in to prevent breaking users’ suspension of disbelief. Users of a world set in Edo-period Japan, for example, would not be amused if an astronaut were to interrupt their immersive experience. Hence, in the case of avatar interoperability, each virtual world would need to figure out specific permissions.

2.1.4. Social graph interoperability

Related to users’ identity is their social graph, which describes the user’s network, including its connections and the nature of those relationships (friend, follower, etc.).¹⁰³ If key aspects of a user’s social graph follow them around, the value of their network is (partially) preserved even when switching worlds. By contrast, having to rebuild your contact list in every virtual world that makes up the metaverse would significantly limit exploration. A continuous social graph would be facilitated by the universal identification system discussed above – it could even be part of it.

As social graph interoperability promotes competition, incumbents are not necessarily interested in providing it. Meta, for example, initially opened up its social graph to developers, among others via the ‘Find Friends’ API. This allowed end-users to find their Facebook connections on other apps. When Twitter’s short-form video app Vine became popular, however, Meta cut off its API access.¹⁰⁴ More generally, access to the API became

94 Platform holders could do little to prevent this, as Discord could be run on a second screen (e.g., a smartphone while playing on a console).

95 Samuel Axon, ‘PlayStation’s new Discord integration is a key step for the cross-play dream’ (*Ars Technica*, 10 March 2023), available at: <https://arstechnica.com/gaming/2023/03/playstations-new-discord-integration-is-a-key-step-for-the-cross-play-dream/>.

96 The interoperability means that players can communicate across platforms playing the same game.

97 In addition, misbehaving users could be more easily and permanently banned from (certain) virtual worlds, see Au (fn 2) 220–21. At the same time, this may require some connection to real-world identities, which can be problematic in repressive regimes. These issues are outside the scope of this paper.

98 The feature itself promotes competition by facilitating multi-homing, see CMA, ‘Online platforms and digital advertising market study’ (Final Report, July 2020) para 3.217. The concerns relate to the way in which the features are provided.

99 Bundeskartellamt, Case B6-55/20, *Meta*, 23 November 2022 (case summary).

100 DMA, Art. 5(7).

101 Ibid Art. 5(2).

102 As such, the independent player is likely less interested in disintermediating business users from end-users, which is a concern for many business users.

103 See CMA, (fn 98) app J, Box 2. For a definition, see LEAR, ‘Ex-post assessment of merger control decisions in digital markets’ (Final Report, May 2019) 60 (‘the global mapping of users which reveals how they are related to each other and what are their preferences and the preferences of the other users they interact with’).

104 Ibid CMA para 34; U.S. House of Representatives (Subcommittee on Antitrust), ‘Investigation of competition in digital markets’ (Majority Staff Report and Recommendations, July 2022) 138–41.

conditioned on not competing with Meta – a policy that the Federal Trade Commission considers anticompetitive.¹⁰⁵ Meta's history also shows the privacy and security risk involved: it was through a social graph API that Cambridge Analytica managed to collect the data not only of the 270,000 users of the *thisisyourdigitallife* app, but also of millions of their connections.¹⁰⁶

2.1.5. Asset interoperability

The interoperability of items (or assets) is an attractive metaverse dream. It lies beyond the realities of communications, identity, and social graph interoperability but is not as far off as interoperability between virtual worlds. The idea is that users are able to purchase an item in one world – say, a Spider-Man 'skin' (avatar costume), Birkin bag, or magical sword – and then take it to another world.¹⁰⁷ Interoperability can expand the market for such items, which would benefit their creators. As for identity and social graph interoperability, there is no legal basis,¹⁰⁸ but firms are taking the first steps toward asset interoperability. After Fortnite maker Epic Games acquired Rocket League, for example, it allowed cross-ownership of cars between the two worlds.¹⁰⁹ Independent firms, such as Lamina1, are trying to realize wider asset interoperability, often based on blockchain technology (including non-fungible tokens [NFTs]).¹¹⁰ The dream is unlikely to be realized any time soon. Again, the challenges are technical and social. Each asset would need to be encoded with detailed physics to allow for logical integration in other worlds, which would themselves need to be designed to accept the transplant. Stephenson, who co-founded Lamina1, calls the idea of dragging and dropping assets from one virtual world to another 'technically ridiculous'.¹¹¹ Moreover, virtual world developers, who have toiled on their specific systems and aesthetics, may not want users to drop in with transplants.

2.1.6. Currency interoperability

Developed virtual worlds tend to have their own currency, such as Roblox's Robux and Fortnite's V-Bucks. When those worlds allow users to cash out their in-world currency, the resulting fiat currency (e.g., USD or EUR) serves as the interoperability layer, as it can be exchanged for the currency of another game.¹¹² Such economic integration with the real world is crucial for the metaverse.¹¹³ In the absence of cashing-out mechanisms, interoperability – i.e., tradability between virtual world currencies at a certain exchange rate – may be helpful. And whether cashing out is allowed, currency interoperability supports the metaverse vision of seamlessly transitioning between different worlds.

There are, however, significant economic hurdles to such interoperability. A user could buy its currency in one virtual world, enriching its owner, to then take it to another virtual world where they spend their time and purchase user-created items, which – depending on the economic arrangements – the owner may not receive any money. Worse, if the creators cash out, the owner actually owes them money. Specific arrangements would need to be made to prevent such imbalances. We have seen a basic version of such an arrangement before. When Sony allowed cross-play of Fortnite on its PlayStation, it wanted to make sure that gamers did not play on its platform while paying on other platforms (call it the 'play here, pay elsewhere' problem). Hence, Sony required that Epic Games make up the difference when its revenue share from Fortnite dropped below 85% of its gameplay share.¹¹⁴ The complexity of such arrangements increases exponentially with the number of virtual worlds involved.

2.1.7. Conclusion

In conclusion, some valuable forms of interoperability – communications, identity, and social graph – are feasible

105 The FTC is fighting the policy in court. The count survived the motion-to-dismiss stage because it also included Meta's acquisitions, but the judge was ready to summarily dismiss it, see *FTC v. Meta Platforms, Inc.*, No. 1:20-cv-03590-JEB (D.D.C. 2022) 34–40.

106 Nathaniel Fruchter, Michael Specter and Ben Yuan, 'Facebook/Cambridge Analytica: lessons and a way forward' (*MIT Internet Policy Research Initiative*, 20 March 2018), available at: <https://internetpolicy.mit.edu/blog-2018-fb-cambridgeanalytica/>.

107 Item interoperability raises IP questions, see Konstantinos Stylianou et al., 'European Commission call on competition in virtual worlds' (CREATe Contribution, 2024) 6–7, but those are beyond the scope of this contribution.

108 Regulation (EU) 2023/2854 of the European Parliament and of the Council on harmonized rules on fair access to and use of data (Data Act), Art. 35(1)(b) concerns the portability of digital assets, but only between cloud computing providers.

109 'Use items across Fortnite and Rocket League with cross-game ownership!' (*Fortnite News*, 8 December 2023), available at: <https://fortnite.com/news/use-items-across-fortnite-and-rocket-league-with-cross-game-ownership>.

com/news/use-items-across-fortnite-and-rocket-league-with-cross-game-ownership.

110 Lamina1, 'Create content on Lamina1' (*Medium*, 18 May 2024), available at: <https://medium.com/@LAMINA1/create-content-on-lamina1-e0b51fc35c6b>.

111 Matthew Ball, 'Interviewing epic games founder/CEO Tim Sweeney and author/entrepreneur Neal Stephenson' (16 July 2024), available at: <https://matthewball.co/all/sweeneystephenson>.

112 Virtual worlds that allow cashing out often set a minimum threshold for doing so.

113 See Section 1.2.

114 Tom Warren, 'Sony really hated PS4 crossplay, confidential documents reveal' (*The Verge*, 3 May 2021), available at: <https://theverge.com/2021/5/3/22417560/>. Similar disputes may occur under the DMA: gatekeepers must allow end-user to access content not purchased on their platform but may like to be compensated for the perusal of that content on their platform.

but require guardrails. Firms have already provided a proof-of-concept, if not more. Competition law can be supportive, though its applicability is triggered mainly when interoperability is first allowed and then denied ('open early, closed late'), as in the case of Meta,¹¹⁵ or in the event of a merger, as in the case of AOL.¹¹⁶ The DMA goes a step further but is only applicable to a limited set of gatekeeper services. Other forms of interoperability are trickier. Currency interoperability faces economic hurdles but is not highly needed in virtual worlds with a cashing-out option. Asset interoperability would bring us a significant step closer to the metaverse but is at present, with very limited exceptions, technologically impossible. Even if it becomes possible, it is likely only desirable in select virtual worlds: those built with interoperability in mind and those centered around broader social – not specific game – mechanics.¹¹⁷

The end-state of the metaverse, with interoperability between virtual worlds, is a long time off. In the early stages, competition policy should not (and cannot) accelerate that future.¹¹⁸ User preferences, and their satisfaction by developers, should be the main driving force. For the metaverse to be compelling to users, it needs to be an improvement over the current situation, where devices (e.g., the iPhone) and platforms (e.g., the PC gaming platform Steam) already allow users to dive into a variety of different worlds.¹¹⁹ How is the metaverse's idea of teleporting between worlds different from closing one app or game and opening another? It can be hard to picture a significant value-add but, as Steve Jobs said, 'people don't know what they want until you show it to them.'¹²⁰

Visionary developers may be able to satisfy an interoperability need users did not know they had. However, are they sufficiently incentivized to provide it? We have seen

interoperability drives before, for example with audio file formats, which went from annoyingly proprietary to seamlessly standardized.¹²¹ It must be noted, however, that user preferences with regard to audio file formats are nowhere as heterogeneous as with regard to virtual worlds. So, even if developers buy into the interoperable metaverse vision, how do they implement it without losing the uniqueness of each virtual world? This requires finding the right level at which to implement interoperability. It may be at the infrastructural level of game engines (or more broadly, graphic engines). For immersive virtual worlds, there is already some convergence on Epic Games' Unreal Engine.

Standard-developing organizations (SDOs) are starting metaverse-specific work, cautious not to be too prescriptive. An alliance has formed behind Universal Scene Description (USD).¹²² Invented and open-sourced by Pixar, it provides a standardized way to animate 3D scenes. NVIDIA's founder Jensen Huang has called USD 'the HTML of the metaverse', referring to Hypertext Markup Language, the standard for the display of web-pages.¹²³ SDOs are in turn meeting in the Metaverse Standards Forum, which has working groups around themes like avatar and asset interoperability.¹²⁴

The fact that competition policy should not (yet) actively work toward a single metaverse, does not mean it should stand by passively either. If a metaverse platform grows large by interconnecting with different virtual worlds but then cuts off access ('open early, closed late'), this presents a strong case for intervention.¹²⁵ Competition law can also come in when parties abuse standard-setting procedures.¹²⁶ If a party influences the standard selection process while not disclosing they hold relevant patents and subsequently enforcing them,¹²⁷ or if

115 Case C-233/23 *Alphabet et al. v. Autorità Garante della Concorrenza e del Mercato ('Android Auto')* EU:C:2025:110 suggests that the EU refusal to supply doctrine may be relied on more generally for access to platforms, at least if (i) a (different) category of apps has already been granted access and (ii) the app of the developer requesting access is in (potential) competition with that of the platform.

116 See also the interoperability obligations imposed in *Google/Fitbit* (Case M.9660) Commission Decision of 17 December 2020 and *Meta/Kustomer* (Case M. 10262) Commission Decision of 27 January 2022.

117 Roblox comes to mind as an example of such a world. In fact, Roblox can already be conceived of as a network of virtual worlds (i.e., the experiences within it). With MSquared, the firm Improbable is also working toward an infrastructure for connected virtual worlds, see <https://msquared.io/>.

118 As shown above, unilateral conduct rules cannot be relied on to bring about such wider interoperability. State aid provides more scope to incentivize the adoption of interoperability, see, e.g., Mario Draghi, 'The future of European competitiveness – part B: in-depth analysis and recommendations' (Report, September 2024) 302.

119 Such multi-homing attenuates network effects and thus weakens the argument for interoperability (see Section 2.1.1). For background, see Jonathan Nuechterlein and Howard Shelanski, 'Lessons from telecom regulation for tech competition policy' (2025) 9–11, available at: <https://ssrn.com/abstract=5172586>.

120 CNBC, 'Steve Jobs 1997 interview: defending his commitment to Apple' (*YouTube*, 27 April 2018), available at: <https://youtube.com/watch?v=xchYT9wz5hk>.

121 The drive was double: audio files were increasingly encoded in a standardized format, while media players increasingly accepted differently encoded files. For an insight into how this played out in the 2000s, see *Microsoft* (Case COMP/C-3/37.792) Commission Decision of 24 March 2004, paras 112–43.

122 See <https://aousd.org/>.

123 Rick Merritt, 'At SIGGRAPH, NVIDIA CEO Jensen Huang illuminates three forces sparking graphics revolution' (*NVIDIA Blog*, 9 August 2022), available at: <https://blogs.nvidia.com/blog/siggraph-huang-metaverse-ai/>.

124 See <https://metaverse-standards.org/>.

125 Generally, see Friso Bostoen and Nicolas Petit, 'Antitrust rules and remedies against platforms' Treacherous Turns' [2025] *European Law Journal* (advance access).

126 Extensively, see Björn Lundqvist, *Standardization under EU Competition Rules and US Antitrust Laws: The Rise and Limits of Self-Regulation* (Edward Elgar, 2014).

127 *Rambus* (Case COMP/38.636) Commission Decision of 9 December 2009.

a holder of a standard essential patent seeks an injunction without first offering fair, reasonable, and non-discriminatory (FRAND) licensing terms,¹²⁸ they may be abusing their dominance.

Before thinking ahead to interoperability, however, the economics of the metaverse must make sense, as I discuss in the next section.

2.2. Mobile distribution

A thriving creator economy is crucial for any metaverse platform.¹²⁹ In presenting Meta's metaverse ambition, Zuckerberg and colleagues referred to creators no less than 50 times.¹³⁰ And firms are putting their money where their mouth is. Meta immediately set up a one billion USD fund to invest in creators.¹³¹ Epic Games has started paying 40% of its net Fortnite revenue to creators based on user engagement with their experiences ('maps').¹³² And as mentioned above, Roblox paid out over 920 million USD to its creators in 2024.¹³³ The priority of this cost becomes particularly clear when you consider that the firm was over one billion USD in the red that year.¹³⁴

Metaverse platforms do not have full control over their creator policies and payments, as they are at the behest of other platforms to reach users. Those platforms can be console- or PC-based, but these days the main interface is mobile. Take Roblox: 80% of its users access the platform on mobile, compared to 20% on PC and console.¹³⁵ In mobile, Apple's iOS rules the day, with Google Android a distant second, at least when measured in consumer spending.¹³⁶ Apple's position is unique in that it stretches across the value chain, from the underlying infrastructure to the user interface. iOS itself can be conceived of as a metaverse platform of sorts, as it is the jump-off point for users to dive into different virtual worlds (apps). Proto-metaverses like Roblox and Fortnite are just virtual

worlds within Apple's system, even if they in turn provide users with a collection of virtual worlds. That is why they are particularly vulnerable to Apple's policies. In financial statements, Roblox warns investors that it 'depend[s] on effectively operating with third-party operating systems, hardware, and networks that may make changes affecting our operating costs, as well as our ability to maintain our Platform'.¹³⁷

The platform threat is twofold. The first issue concerns the fee levied on metaverse payments. Apple imposes a 30% App Store fee on virtual world transactions carried out within iOS.¹³⁸ This can seriously reduce the rewards for user creation. A metaverse platform has to pay for its infrastructure, R&D, moderation, marketing, and other expenses. Whatever revenue remains can be shared with creators but, especially after subtracting another 30%, little may be left. This limits creation, the main driver of the metaverse. A second issue concerns policies on what is allowed within iOS. Apple can prohibit certain types of creation or impede supporting technologies. The two issues can lead to similar results. Apple initially prohibited the sale of NFTs; later, it allowed NFT trading but subjected it to its 30% fee.¹³⁹ Under the new policy, it remains unfeasible to trade NFTs, but now as a matter of economics rather than policy – compare it to trading stocks subject to a 30% fee, which would make no economic sense either.

Let us unpack each issue, starting with restrictive App Store policies. In theory, these are the biggest threats to metaverse development. The most important App Store rule is that developers may not create an app that serves as 'an interface for displaying third-party apps'.¹⁴⁰ This rule is intended to prevent other app stores on iOS, as Apple wishes to reserve the monopoly on app distribution to its own App Store, but the rule reaches further. Its extent became more apparent when Apple held that

128 Case C-170/13, *Huawei Technologies Co. Ltd. v. ZTE Corp. and ZTE Deutschland GmbH* EU:C:2015:477.

129 See Au (fn 2) 14–16 on how the metaverse 'has powerful experience creation tools' and 'is integrated with the real-world economy'; Ball (fn 1) 25 on how the metaverse requires 'an open economy of virtual goods'.

130 Zuckerberg (fn 27).

131 'Investing \$1 billion in creators' (*Meta Newsroom*, 14 July 2021), available at: <https://about.fb.com/news/2021/07/investing-1-billion-dollars-in-creators/>. More recently, see 'Empowering Meta Horizon creators with a new \$50M fund and advanced tools' (*Meta Horizon Blog*, 20 February 2025), available at: <https://developers.meta.com/horizon/blog/empowering-horizon-creators-with-a-new-50m-fund-and-advanced-tools-for-mobile-and-mr-worlds>. See similarly Rita Liao, 'South Korea's Naver Z launches \$100M fund for metaverse creators' (*TechCrunch*, 27 January 2022), available at: <https://techcrunch.com/2022/01/27/south-koreas-naver-z-launches-100m-fund-for-metaverse-creators/>.

132 'Introducing creator economy 2.0' (*Epic Games Creator Portal*, 22 March 2023), available at: <https://create.fortnite.com/news/introducing-the-creator-economy-2-0> (note that revenue is distributed to independent

creators and Epic Games itself, e.g., for engagement with its popular *Battle Royale* map).

133 Fortnite 350 million USD, see Section 1.2.

134 Roblox Corporation (fn 23) 77. Its only greater cost center was R&D.

135 Ibid 9. More (hardcore) gaming-focused proto-metaverses see a lower share of mobile players, due to the need for effective controls, which mobile lacks (unless an external controller is used).

136 Ibid 28. Two-thirds of Roblox's mobile revenue comes from Apple and one-third from Google, which is in line with the respective app stores' revenue more generally, see Laura Ceci, 'Global Google Play and Apple App Store app revenues 2024' (*Statista*, 30 August 2024), available at: <https://statista.com/statistics/183469/app-stores-global-revenues/>.

137 Roblox Corporation (fn 23) 28–29.

138 There are certain exceptions to the 30% fee, though they tend not to apply to the virtual world transactions discussed here.

139 For App Store Review Guidelines (s. 3.1.1), see <https://developer.apple.com/app-store/review/guidelines/>.

140 Ibid s. 3.2.2(i).

cloud-based subscription gaming apps fall under the prohibition.¹⁴¹ Cloud gaming services provide users access to a variety of games through a single app – a ‘Netflix for games’, if you will.¹⁴² Apple later nuanced its ban, ‘allowing’ cloud gaming as long as every single game was downloaded directly from the App Store.¹⁴³ As with NFTs, the new policy still constitutes a quasi-ban – imagine having to separately download every individual Netflix movie/episode to your device before viewing.¹⁴⁴

By prohibiting cloud gaming apps, Sweeney argued that Apple ‘outlawed the metaverse’.¹⁴⁵ After all, the metaverse is a portal to various virtual worlds, often games; in other words, an app serving as an interface for other apps. Interpreted strictly, apps like Roblox are non-compliant with the rule. This became painfully clear during the *Epic v. Apple* trial, where the parties debated whether Roblox contained user-created games, in which case it should be banned from the App Store.¹⁴⁶ Apple’s witness demurred, arguing that the virtual worlds within Roblox are better qualified as ‘experiences’ than ‘games’. Even though the witness was not able to convince the judge, Roblox quickly moved to align reality with Apple’s conception of it, renaming its user-created virtual worlds from ‘games’ to ‘experiences’.¹⁴⁷ Later, Apple retroactively legitimized Roblox by adding a new App Store guideline that allows apps displaying ‘user-generated experiences’.¹⁴⁸

Apple’s reluctance to accommodate (proto-) metaverses is a matter of incentives. It does not want developers to create experiences for Roblox, which disintermediates Apple from both the developer and the end-user; it wants developers to create apps for iOS. In its mobile ecosystem market study, the UK Competition & Markets Authority (CMA) took a closer look at Apple’s incentives to ban cloud gaming, which are informative.¹⁴⁹

First, Apple wants to retain its lucrative app distribution monopoly, which could be undermined by the alternative distribution model of cloud gaming. Second, Apple wants to protect the importance of its high-end devices (and the associated revenue). Unlike on-device gaming, cloud gaming does not rely on premium hardware for its performance. Hence, cloud gaming can diminish the importance of premium devices such as Apple’s, leading to lower sales.¹⁵⁰

The incentives to ban the metaverse entirely are slightly different. The first incentive, to retain an app distribution monopoly, certainly applies. In fact, there may be a stronger incentive: if the metaverse is indeed the successor to mobile, then the metaverse does not just chip away at mobile app distribution but displaces mobile ecosystems entirely. As the leader in mobile, Apple has a strong interest in preventing such a platform shift. The second incentive is more ambiguous. The metaverse is computationally intensive, which means Apple’s premium hardware is a good fit. It could even drive device sales, at least as long as processing takes place on-device rather than in the cloud.

Let us move then from the first issue of mobile distribution *policies* to the second issue of mobile distribution *payments*. The metaverse is estimated to contribute anywhere between 1.3 and 16 trillion USD to global GDP by 2030.¹⁵¹ NVIDIA’s Huang has predicted the metaverse economy will be larger than the physical economy, in part because ‘creators will make more things in virtual worlds than they do in the physical world’.¹⁵² Even though such estimates should be taken with a grain (or even a whole pinch) of salt, the metaverse could certainly host significant economic activity. If mobile continues to be the main access point, then Apple is likely to charge a

141 Nick Statt, ‘Apple confirms cloud gaming services like xCloud and Stadia violate App Store guidelines’ (*The Verge*, 6 August 2020), available at: <https://theverge.com/2020/8/6/21357771/>.

142 For the sake of simplicity, I refer to ‘cloud gaming’ as a shorthand for ‘cloud-based subscription gaming’. Cloud gaming encompasses a variety of game distribution modes, including such ‘cloud gaming as a feature’, see Joost Rietveld, ‘Cloud gaming is not a distinct market: a typology of cloud gaming services and what it means for Microsoft’s proposed acquisition of Activision Blizzard’ (Submission to the CMA, 31 March 2023).

143 App Store Review Guidelines (fn 139) s. 3.1.2(a).

144 See similarly CMA, ‘Mobile ecosystems market study’ (Final Report, June 2022) para 6.226–6.231 (on how Apple continues to make cloud gaming infeasible), and Alex Heath, ‘Facebook brings cloud gaming to Apple devices with a web app’ (*The Verge*, 23 July 2021), available at: <https://theverge.com/2021/7/23/22589398/> (quoting Meta’s vice president of gaming as saying, ‘Apple’s policy to “allow” cloud games on the App Store doesn’t allow for much at all’).

145 Tim Sweeney (*X*, 7 August 2020), available at: <https://x.com/TimSweeneyEpic/status/1291509151567425536>.

146 *Epic Games, Inc v. Apple Inc.*, 559 F. Supp. 3d 898 (N.D. Cal. 2021) 67–70 and in particular fns 354–55. An earlier internal email shows surprise within Apple that they even approved Roblox for inclusion in the App Store,

given that it presents an interface for apps, see Exhibit No. PX-0305, available at: <https://embed.documentcloud.org/documents/21043939-2017-july-contradiction-roblox-approved-despite-store-within-store-trystan-read-about-it/>.

147 Adi Robertson, ‘Apple said roblox developers don’t make games, and now Roblox agrees’ (*The Verge*, 14 May 2021), available at: <https://theverge.com/2021/5/14/22436014/>.

148 App Store Review Guidelines (fn 139) s. 1.2.1. It is doubtful whether Roblox even falls under this guideline, which refers to ‘non-developer creators’, as Roblox hosts many professional developers.

149 CMA, (fn 144) paras 6.236–6.252.

150 Relatedly, cloud gaming – as a device-agnostic service – can reduce switching costs between devices.

151 See Ball (fn 1) 26 collecting estimates of McKinsey, KPMG, etc. For another study, see Lau Christensen and Alex Robinson, ‘The potential global economic impact of the metaverse’ (Study) (2022), available at: <https://analysisgroup.com/globalassets/insights/publishing/2022-the-potential-global-economic-impact-of-the-metaverse.pdf>.

152 Jacob Roach, ‘Nvidia CEO says the metaverse will be “larger than the physical world”’ (*digitaltrends*, 10 November 2021) <https://digitaltrends.com/computing/nvidia-ceo-metaverse-larger-than-real-world/>.

30% fee for that economic activity. It already charges in-game purchases. The metaverse hosts a wider variety of experiences, but it is clear that whenever activity moves online – say, gym classes during the pandemic – Apple wants a cut.¹⁵³

The 30% commission far exceeds the fees charged by credit cards (e.g., Visa) and online payment systems (e.g., PayPal), both of which charge around 3%. The reasons are partly historical.¹⁵⁴ When Nintendo started allowing third parties to develop games for its console, it charged them a 30% fee: 10% for licensing and 20% for cartridge manufacturing. The 30% fee was adopted by other console platforms (Xbox and PlayStation), PC game platforms (in particular Steam), and finally Apple. Along the way, cartridges disappeared, but the 20% charged for manufacturing them endured. As a justification, console manufacturers can refer, with some credibility, to the fact that they sell their hardware at a loss and need to subsidize it with commission fees on third-party software sales.¹⁵⁵ This is not the case for PC game platforms, where the 30% fee is stuck because of Valve's quasi-monopoly (indeed, when Epic Games launched its own store, the fees dropped).¹⁵⁶ Monopoly power over app distribution similarly explains a significant part of Apple's fee.

So, virtual world economies are already large but may become a lot larger still; payments within them are, however, a lot more expensive than in the real world. Apple's supracompetitive fee of 30% on virtual transactions is an important reason. At the same time, it must be recognized that the fee represents more than simple payment processing: Apple provides developers with the iOS software development kit (SDK), which includes access to its APIs, and it moderates its ecosystem, in particular by reviewing every submitted app. And there may also

be rent-seeking elsewhere in the value chain. The game engine Unity, for example, caused a stir when it introduced a 'runtime fee' that would increase developers' costs. However, no cost – except for the one charged by the virtual world itself – is quite as large as the App Store fee.¹⁵⁷

The problem is not so much the commission fee as such but rather its cumulative nature. At present, virtual worlds are not an alternative to iOS/the App Store; rather, they exist within it. This means that a creator owes a commission fee to both the operator of the virtual world and the operating system – a problem of 'double marginalization'.¹⁵⁸ Epic Games prevents the addition of profit margins on PCs by distributing Fortnite through its own store.¹⁵⁹ Roblox, a mobile-first app, does not have that option. It provides a good overview of how the fees add up: for every dollar spent in Roblox, the firm

- pays 22% in app store and payment processing fees (this is slightly lower than Apple's and Google's 30% fee because sales via PC are not subject to it,¹⁶⁰ which brings down the average cost);
- uses 18% to cover platform hosting, i.e., server maintenance, as well as customer support, moderation, user acquisition, translation, and local compliance;
- invests 15% in its platform, e.g., the development of better creator tools.

What remains is split between Roblox (17%, to cover operating costs) and creators (28%).¹⁶¹ Creators thus walk away with less than 30% of every dollar spent on their experiences, which means they make less from their experiences than Apple.¹⁶² Roblox takes an altogether greater cut than Apple, but it provides significantly more

153 Jack Nicas and David McCabe, 'Their businesses went virtual. Then Apple wanted a cut' *New York Times* (28 July 2020), available at: <https://nytimes.com/2020/07/28/technology/apple-app-store-airbnb-class-pass.html>. Other creator platforms that Apple is requiring to pay the 30% fee include Patreon and Substack, see 'iOS in-app purchases FAQ' (*Patreon Help Center*, 9 September 2024), available at: <https://support.patreon.com/hc/en-us/articles/27992151772813-iOS-in-app-purchases-FAQ> and Hamish McKenzie, 'The price of payments' (*On Substack*, 12 August 2024), available at: <https://on.substack.com/p/apple-patreon-in-app-purchase-system>.

154 Takashi Mochizuki and Vlad Savov, 'Epic's battle with Apple and Google has roots in the pac-man era' (*Bloomberg*, 19 August 2020), available at: <https://news.bloomberglaw.com/antitrust/epics-battle-with-apple-and-google-has-roots-in-the-pac-man-era>.

155 See (fn 146) 78–79. Nintendo, which tends not to make a loss on its hardware, cannot rely on this justification. Additional reasons that the 30% fee may be more justified on consoles include the platform holders' greater investment in content (first- and third-party), better curation and discovery, and stricter review. See the remarks of Rick Gibson at 'My Gaming Device is mine: is big gaming letting us play?' (*Euroconsumers 'Start Talking' Webinar*, 25 April 2024), available at: <https://euroconsumers.org/webinars/5578/>.

156 Making that case, see *Wolfire Games LLC v. Valve Corp.*, Second Amended Complaint, No. 2:21-cv-00563-JCC (W.D. Wash. 2023) [on Epic's entry specifically, see (nn 79–81)].

157 Costs are lower elsewhere due to the availability of alternatives. Unity's runtime fee led creators to (threaten to) swap its game engine for a competitor; eventually, Unity cancelled the new fee, see <https://unity.com/blog/unity-is-cancelling-the-runtime-fee>.

158 See Philippe Choné, Laurent Linnemer and Thibaud Vergé, 'Double marginalization and vertical integration' (2021) CESifo Working Paper No. 8971.

159 Providing context on how the Epic Games Store was a reaction to Steam's high fees, see (fn 156) Exhibit 15, available at: <https://drive.google.com/file/d/1a-HLEOqbg7QqUemQv0YyunnxI7lN03w1/view>.

160 Roblox does not make its app available on Steam; it must be downloaded directly on PC.

161 See 'Earning on Roblox' (*Roblox Creator Hub*), available at: <https://create.roblox.com/docs/production/earning-on-roblox>. The 28% developer share consists of 25% from cashing out Robux and 3% engagement-based payouts.

162 At least from transactions taking place on mobile, which is the large majority.

services, including servers, creator tools, multiplayer networking, and user acquisition.¹⁶³ From the point of view of Roblox creators, Apple just provides payment processing.¹⁶⁴ The difference in services between Roblox and Apple is reflected in their respective profit margins. Roblox's operating loss of over one billion USD represents a –30% profit margin, which is in no small part due to the 40% of revenue it spends on research and development (R&D).¹⁶⁵ Apple's App Store has an operating margin of 75%–80%, with little R&D.¹⁶⁶

Competition law is not there to divide profit margins,¹⁶⁷ though it may come in when one firm affects another firm's margins in a way that distorts competition.¹⁶⁸ When it comes to metaverse platforms, this could happen in two ways. First, high mobile distribution fees can make it exceedingly difficult for a creator-led virtual world to turn a profit. In the long term, that leads existing virtual worlds to shut down; others do not even get off the ground. App stores, which stand to be replaced by such virtual worlds, might not mind such an outcome. At the same time, it is difficult to argue app stores actively pursue that outcome, as their commission fees were set long before.¹⁶⁹ Second and relatedly, high mobile distribution fees depress the remuneration received by creators.¹⁷⁰ Logically, their incentive to create is thus diminished¹⁷¹

(even if the incentives of creators are not purely monetary).¹⁷² In turn, reduced creative incentives stunt the growth of virtual world economies.¹⁷³ Any anticompetitive goal on the part of app stores would be aligned with the one discussed above: avoiding disintermediation by virtual worlds by making it more attractive for creators to develop apps instead of virtual world experiences.¹⁷⁴ At the same time, app stores could always point to justifications for their high fees, such as the need to monetize their IP.¹⁷⁵

In conclusion, mobile distribution policies can directly impede the metaverse, at least as long as smartphones remain the primary interface. App store payment rules can also slow down the development of the metaverse but do so indirectly, by stunting the growth of the creator economy. Given the economic potential of the metaverse, the wider economy may be harmed. In the last few years, there has been strong pushback against restrictive mobile distribution policies and high app store fees, sometimes with the metaverse explicitly in mind.

Epic Games gave the starting shot by suing Apple in a U.S. court.¹⁷⁶ It argued that Apple's rules against competing app stores and alternative payment mechanisms were a form of monopolization that blocked Epic's metaverse ambitions. In the end, the judge dismissed most of the

163 See the remarks of Joe Ferencz, founder of Gamefam (operator of the largest portfolio of titles on Roblox) on 'Understanding the Roblox creator ecosystem' (*Mobile Dev Memo Podcast*, 6 August 2024), available at: <https://mobiledevmemo.com/podcast-understanding-the-roblox-creator-ecosystem-with-joe-ferencz/> (on how they need to back-end engineers because Roblox takes care of everything).

164 The iOS SDK is mainly relevant for Roblox itself (even then, likely to a limited extent), and app review of the Roblox app in its entirety only occurs once (updates are also approved, but with an established app the review is likely minimal).

165 Roblox Corporation (fn 23) 77.

166 Apple's financial statements do not break out App Store margins and R&D; see (fn 146) 41–43 and 102.

167 An exception is found in mechanisms forcing platforms to bargain with publishers. In many countries, the mechanism is based on IP law (even if sometimes enforced via competition law), but in some countries, like Australia, the media bargaining is part of competition policy. Discussing both the IP and competition-based regimes, see Ula Furgal, 'The Emperor has no clothes: how the press publishers' right implementation exposes its shortcomings' (2023) 72 GRUR International 650. Bargaining mechanisms could also be imposed via platform regulation. The CMA is currently investigating whether Google's search services have 'strategic market status' under the Digital Markets, Competition and Consumers Act (DMCCA) and has organized a roundtable with publishers, which may inform its subsequent regulatory approach, see CMA, 'SMS investigation into Google's general search and search advertising services', available at: <https://gov.uk/cma-cases/sms-investigation-into-googles-general-search-and-search-advertising-services>.

168 The most obvious example being margin squeeze, see Friso Bostoën, 'Online platforms and vertical integration: the return of margin squeeze?' (2018) 6 Journal of Antitrust Enforcement 355.

169 Hence, the conduct consists in keeping the fees high while they should have gone down over time.

170 Making that connection, see Ina Fried, 'Roblox CEO: lower app store commissions would benefit developers' (*Axios*, 7 June 2021), available at:

<https://axios.com/2021/06/07/roblox-ceo-lower-app-store-commissions> (on how if app stores took less in fees, Roblox could pass 'the majority of that' money back to developers).

171 Voicing that logic, see Enrico D'Angelo, Roblox's vice president of economy, on 'Building the creator economy, part 1: Roblox' (*Unboxing*, 18 October 2023), available at: <https://unboxingpodcast.substack.com/p/building-the-creator-economy-part> ('if [creators] make no money from building [experiences], nobody will build them, nobody will invest, nobody will create a team to continue building and evolving these applications'). Note that there are also non-economic incentives to create.

172 See, e.g., Takahashi (fn 17): 'Roblox noted that a recent survey of the developer community found that the primary motivator for most Roblox creators is not monetary gain. When asked about what motivates them to develop experiences on Roblox, having fun ranked four times higher than monetization'.

173 See, in that sense, Ben Thompson, 'An interview with Mark Zuckerberg about the metaverse' (*Stratechery*, 28 October 2021), available at: <https://stratechery.com/2021/an-interview-with-mark-zuckerberg-about-the-metaverse/> ('the best way to increase the GDP of the metaverse is to have the fees be as low as possible and as favorable as possible to creators').

174 See, similarly, *Apple – App Store Practices (Music Streaming)* (Case AT.40437) Commission Decision of 4 March 2024, para 343 ('Apple has no incentive to allow alternative distribution methods of native apps that are not fully controlled by Apple and that would remove the "exclusive" position Apple holds as an intermediary between developers and consumers').

175 See (fn 146) 104–14. Monetization is also important for investments into the quality and security of the ecosystem. The need for a specific level of monetization may, however, be nuanced by the app stores' profit margins and R&D expenses (fn 166).

176 Spotify had already complained about Apple's App Store policies to the European Commission, and there had been several national cases, but none of those actions related to the metaverse.

claims, imposing only a limited remedy on Apple: it had to allow developers to steer users to alternative payment systems, which may allow them to bypass the 30% fee.¹⁷⁷ Epic was more successful against Google: a jury ruled that it had unlawfully tied its Play Store and its payment system.¹⁷⁸ Senators introduced the Open App Markets Act to regulate mobile distribution more strictly and comprehensively, but the bill did not make it through Congress.¹⁷⁹

In the EU, the European Commission also found Apple's anti-steering provisions illegal as an abuse of dominance, in particular unfair (exploitative) terms and conditions, but only when applied to music streaming services.¹⁸⁰ In parallel, lawmakers regulated app stores as part of the DMA. The DMA's relevant obligations¹⁸¹ only apply to app stores designated with gatekeeper status, that is, the App Store and Google Play.¹⁸² As gatekeepers, Apple and Google have to open up mobile distribution, in particular by allowing competing app stores and direct downloads of apps ('sideloading').¹⁸³ These obligations are meant to create space for other platforms, such as virtual worlds, which may one day displace the gatekeepers.¹⁸⁴ In addition, Apple and Google have to adapt their payment rules: developer fees must be FRAND; in addition, developers must be able to use/steer users toward alternative payment systems.¹⁸⁵ These obligations are meant to prevent the gatekeepers from obtaining a disproportionate advantage in their relations with business users – a category that includes virtual world creators.¹⁸⁶

The effects of app store regulation are showing only slowly. The gatekeepers have launched compliance initiatives, but those tend to retain high fees and other restrictions, also for apps downloaded outside of their app stores.¹⁸⁷ The mobile distribution process has therefore changed little so far.¹⁸⁸ The Commission has already launched non-compliance investigations against both Apple and Google.¹⁸⁹ Lawmakers in other parts of the world are also recognizing the crucial nature of mobile ecosystems, their embedded app stores, and the ways in which they can impede innovation – a recognition that led to app store acts in South Korea and Japan.¹⁹⁰

As the regulatory tide is turning against the gatekeeper app stores, market signals may offer another reason for Apple and Google to rethink their restrictions. Reputation is an important disciplining force, especially for platforms. If a platform operator develops a reputation for exploiting its business users, those users will, over time, stop investing in the platform. Mobile ecosystems are somewhat shielded from this pressure, as app developers have few alternatives to turn to. But even then, developer discontent is starting to manifest. When Apple launched its Vision Pro headset, for example, it was no great success. The device's failure was attributed in part to a lack of support by developers, who had become disillusioned with Apple.¹⁹¹ And when Roblox announced it would allow creators to offer paid access to their experiences, it limited payments to PCs, which means mobile users may miss out.¹⁹²

177 See (fn 146), mostly affirmed in *Epic Games, Inc v. Apple, Inc*, No 21-16506 (9th Cir. 2023).

178 Verdict Form in *Google Play Store Antitrust Litigation*, No. 20-cv-05671-JD (N.D.C. 2023). The judge still has to rule on remedies. One option is to untie the billing system from the Play Store so that developers are entirely free in choosing their billing system.

179 S.2710 – Open App Markets Act, 117th Congress (2021–2022).

180 See (fn 174).

181 For a more in-depth discussion, see Friso Bostoen, 'Understanding the Digital Markets Act' (2023) 68 Antitrust Bulletin 263, 299–303.

182 *Apple – Online Intermediation Services – App Stores* (Case DMA.100013) Commission Decision of 5 September 2023; *Alphabet – OIS App Stores* (Case DMA.100002) Commission Decision of 5 September 2023.

183 DMA, Art. 6(4). In addition, Art. 5(7) obliges gatekeepers to allow competing browser engines. As current browser engines (in particular Safari's WebKit) are underpowered, this obligation could lead to browsers that can more effectively run web apps. See (fn 144) s. 5.

184 The DMA's goal of contestability, defined in recital 32.

185 DMA, Arts 6(12) and 5(4)–(5) plus 5(7), respectively. The FRAND obligation applies not only to fees but also to wider app store policies, which means it may force app stores to (truly) allow cloud gaming apps, see Damien Geradin and Stijn Huijts, 'Dark clouds gather – the development of cloud gaming, and competition agencies' efforts to enable it on mobile app stores' (2023) 6 Interactive Entertainment Law Review 3.

186 The DMA's goal of fairness, defined in recital 33.

187 See Apple, 'Non-confidential summary of DMA compliance report' (7 March 2025), available at: <https://apple.com/legal/dma/NCS-March-2025.pdf>; Alphabet, 'EU Digital Markets Act compliance report non-confidential summary' (7 March 2025), available at: <https://transparencyreport.google.com/report-downloads>.

188 Examining app store distribution policies, and how they might comply with the DMA more effectively, see Fiona Scott Morton et al., 'Access pricing for app stores under the Digital Markets Act' (2024) Yale Tobin Center for Economic Policy – Digital Regulation Project, Policy Discussion Paper No. 9.

189 *Apple New Business Terms* (Case DMA.100206) Commission Decision of 24 June 2024; *Alphabet – Online Intermediation Services – App Stores – Google Play – Article 5(4)* (Case DMA.100075) Commission Decision of 25 March 2023.

190 For South Korea, see the Telecommunications Business Act, Art. 50(8)–11; for Japan, see Simon Vandewalle, 'Japan's new smartphone law', available at: <https://sites.google.com/view/vandewalle/scrabbles/japans-smartphone-law> (including a partial English translation).

191 Ben Thompson, 'The Apple vision pro's missing apps' (*Stratechery*, 22 January 2024), available at: <https://stratechery.com/2024/the-apple-vision-pros-missing-apps/>. Putting numbers on the Vision Pro's failure with developers, see Matthew Ball, '9 takeaways from the vision pro after 6 months' (12 August 2024), available at: <https://matthewball.co/all/applevision2024>.

192 David Baszucki, 'RDC 2024: Roblox's next frontier' (*Roblox Newsroom*, 6 September 2024), available at: <https://corp.roblox.com/newsroom/2024/09/rdc-2024-robloxs-next-frontier>. It seems like users may be able to pay for access on PCs and then access the experience on mobile. The avoidance of mobile payments is likely necessitated by the higher revenue share (of 50%–70%) that Roblox wants to give developers. At the same time, users are now essentially purchasing games in Roblox, which brings it into even closer conflict with Apple's 'no apps in apps' rule.

3. Conclusion

Novel technologies go through ‘hype cycles’.¹⁹³ The metaverse reached a ‘peak of inflated expectations’ around 2022 – and such a peak is inevitably followed by a ‘through of disillusionment’. The policy is not isolated from these hype cycles. This is visible in how the European Commission initially centered its strategy on the metaverse but has meanwhile pivoted to the more down-to-earth concept of virtual worlds. Whatever the terminology, however, virtual worlds are bound to become more immersive and integrated with the real world in the coming years. At some point, we are bound to reach a ‘slope of enlightenment’, where the technology’s potential becomes more tangible and understood. ‘When’ is an open question: Meta itself is unsure whether it will take 5–10 years or rather 10–15 years to realize the metaverse.¹⁹⁴

Even if the metaverse is a futuristic concept – that will remain in the future for some time – we can already discern what it might look like. Science fiction literature gave us the concept, and the proto-metaverses in existence confirm the direction of travel: the metaverse is an immersive, continuous, persistent, synchronous, and massively scaled virtual world – or rather, an interoperable collection of virtual worlds. And those worlds are largely shaped by creators that can thrive economically. Here too, fundamental questions remain unanswered: Is the metaverse the next version of the internet, and if so, will it uphold its inherent openness? Or is the metaverse the platform that displaces mobile, and if so, will it be similarly controlled by one or two gatekeepers?

The answers to these questions will mainly come from developers. However, the law can play a role in steering their efforts, or at least keeping them within certain guardrails. Not every firm necessarily has an incentive to make their virtual worlds interoperate and to let creators thrive (outside of their own walled garden). When it comes to interoperability, competition law enforcement can target situations in which firms initially grant interoperability to accelerate their growth, to then opportunistically cut it off once they are in a position of (market) power. When interoperability is not granted in the first place, however, competition law has little ability to compel it. Procompetitive regulation, like the DMA, is a more appropriate instrument to burden firms with such positive obligations. Even then, virtual world interoperability – insofar as it is desirable – would require a separate regulatory initiative.

The story is similar for mobile distribution policies, including fees, which impede metaverse creation. Competition enforcement law enforcement can make a difference on the margins, e.g., by prohibiting anti-steering provisions that lock developers and users into a specific payment system (with a high fee). Even in such cases, however, competition law remedies have shown little effectiveness. The DMA is potentially more effective here: it is not asked to comprehensively regulate an unknown future, as with interoperability; it only needs to ensure that existing platforms, in particular mobile ecosystems, leave room for undefined innovation. It does so by opening up app distribution, including by forcing closed mobile ecosystems to allow competing app stores. Once gatekeepers comply with those obligations, it is up to developers and creators to do the rest.

193 As tracked by Gartner, see <https://www.gartner.com/en/research/methodologies/gartner-hype-cycle>. Note, however, that the metaverse is not a single technology but a bundle of technologies.

194 Compare Zuckerberg (fn 27) with Andrew Bosworth, ‘Building the metaverse responsibly’ (*Meta Newsroom*, 27 September 2021), available at: <https://about.fb.com/news/2021/09/building-the-metaverse-responsibly/>.